LESSON 01 / 09

What a set is (and what it is not)

A **set** is a well-defined collection of distinct objects. Well-defined means membership is decidable: for any candidate x, the statement $x \in A$ is either true or false, never a matter of taste. The objects are called **elements** (or members). Distinct means a set lists each element once: $\{1, 1, 2\}$ means $\{1, 2\}$.

Notation. Capital letters (A, B, U) name sets, small letters (x, y, a) name elements. $x \in A$ reads "x is an element of A"; $x \notin A$ is its negation. The **cardinality** $n(A)$ (also written $|A|$) is the number of elements; it may be finite or infinite.

Three ways to write a set

- **Roster (listing):** $A = \{1, 2, 3\}$ — order and repetition do not matter.
- **Set-builder:** $B = \{x : x \text{ is an even prime}\} = \{2\}$. Read $\{x : P(x)\}$ as "the set of all x such that $P(x)$ holds".
- **Interval (only for subsets of $\mathbb{R}$, order matters):** (a, b) open, $[a, b]$ closed, $[a, b)$ or $(a, b]$ half-open, and rays $[a, \infty)$, $(-\infty, b)$. Infinity is never an element: always round brackets at $\pm\infty$.

> **ASSUMPTIONS & CONVENTIONS**
> Braces $\{\}$ always describe a set. $\in$ links an **element** to a set; $\subseteq$ links a **set** to a set. $1 \in \{1, 2\}$ but $\{1\} \subseteq \{1, 2\}$. Never mix the two levels: "$1 \subseteq \{1, 2\}$" is a type error, not a false inequality. All sets in a discussion are assumed to live inside a fixed **universal set** U once complements appear.

Kinds of sets

- **Empty (null) set** $\varnothing$: the set with no elements. There is exactly **one** empty set (uniqueness: any two sets with no elements are equal by definition of equality). $n(\varnothing) = 0$.
- **Singleton** $\{a\}$: one element. Note $\{a\} \neq a$ — the set and its element are different kinds of objects.
- **Finite / infinite:** $n(A)$ is a natural number vs not. $n(\mathbb{N})$ is infinite (countably so; no JEE counting applies).
- **Equal sets:** $A = B \Leftrightarrow$ every element of A is in B and vice versa. The standard proof is **double inclusion**: prove $A \subseteq B$ and $B \subseteq A$ separately.
- **Equivalent sets:** same cardinality only. $\{1, 2\}$ and $\{3, 4\}$ are equivalent but not equal. Do not say "equivalent" when you mean "equal".

Subsets and the power set

$A \subseteq B$ iff every element of A is an element of B. We write $A \subset B$ (or $A \subsetneq B$) when $A \subseteq B$ with $A \neq B$ (**proper** subset). Vacuous truth: $\emptyset \subseteq A$ for every A — there is no element of $\emptyset$ that could fail the test.

SHORT DERIVATION

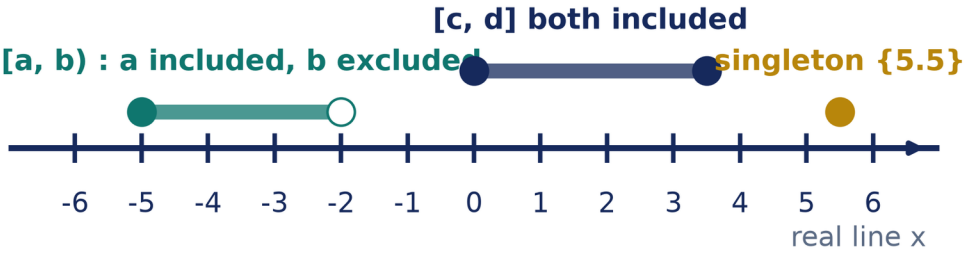
Why $|\mathcal{P}(A)| = 2^n$

Build a subset $S \subseteq A$ by deciding, for each of the n elements of A in turn, "in S" or "not in S". That is 2 choices per element, independently, so $2 \times 2 \times \cdots \times 2 = 2^n$ subsets. Small case $A = \{1, 2\}$: the four decisions list $\emptyset, \{1\}, \{2\}, \{1, 2\}$. Same reasoning: subsets with exactly k elements number $\binom{n}{k}$, so $\sum_{k=0}^{n} \binom{n}{k} = 2^n$ (choosing each element is a subset of some size k).

KEY RESULT

The **power set** $\mathcal{P}(A)$ is the set of all subsets of A. Then $|\mathcal{P}(A)| = 2^{n(A)}$ and, for finite A: number of subsets $= 2^n$, number of proper subsets $= 2^n - 1$, number of non-empty subsets $= 2^n - 1$, number of non-empty proper subsets $= 2^n - 2$. Also $A \in \mathcal{P}(A)$ (as an element) and $A \subseteq A$ (as a subset) — two different statements, both true.

Interval conventions on the real line (filled dot = included)



The four confusions that cost marks

COMMON TRAP

1. $\emptyset$ **vs** $\{0\}$ **vs** $\{\emptyset\}$. $\emptyset$ has 0 elements. $\{0\}$ has one element (the number 0). $\{\emptyset\}$ has one element (the empty set). So $n(\{\emptyset\}) = 1$, and $\emptyset \in \{\emptyset\}$ is true while $\emptyset \in \{0\}$ is false. The set $\{\emptyset, \{0\}\}$ has exactly two elements.
2. $\in$ **vs** $\subseteq$. $\emptyset \subseteq \{0\}$ is true (subset), $\emptyset \in \{0\}$ is false (not an element). $0 \in \{0\}$ true, $\{0\} \in \{0\}$ false.
3. ***Equal vs equivalent.*** $\{1, 2\}$ and $\{2, 1\}$ are equal; $\{1, 2\}$ and $\{3, 4\}$ are only equivalent.
4. ***Two empty sets in one problem*** (e.g. $A \cap B = \emptyset$ vs $A \setminus B = \emptyset$) mean different things: the first says A, B are disjoint, the second says $A \subseteq B$.

WORKED EXAMPLE

Worked. $A = \{x \in \mathbb{Z} : |x| \le 2\}$ and $B = \{x \in \mathbb{Z} : 1 \le x \le 3\}$. Then $A = \{-2, -1, 0, 1, 2\}$, $B = \{1, 2, 3\}$, $A \cap B = \{1, 2\}$, $A \cup B = \{-2, -1, 0, 1, 2, 3\}$, $n(A \cup B) = 7$. **Check:** the two listed sets share exactly $1, 2$, and every integer appearing in either list appears once in the union. Notice $|x| \le 2$ includes $x = -2$ — boundary cases always flip signs when absolute values appear.

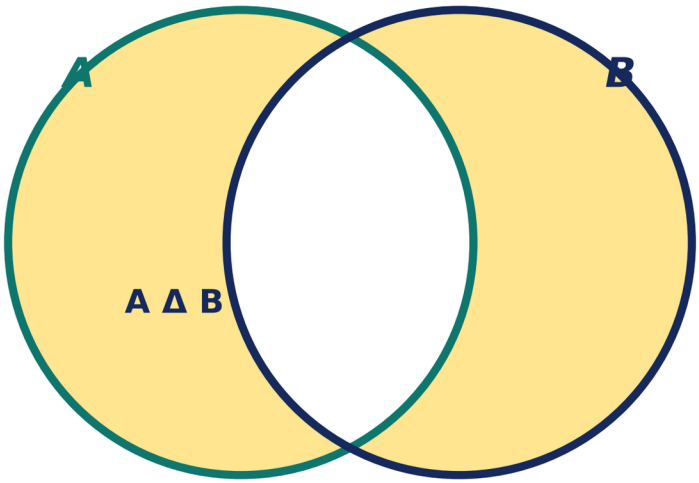
LESSON 02 / 09

The five operations

Fix a universal set U; all complements are relative to U.

- **Union:** $A \cup B = \{x : x \in A \text{ or } x \in B\}$ — "or" is inclusive: $A \cap B$ is counted in.
- **Intersection:** $A \cap B = \{x : x \in A \text{ and } x \in B\}$. Disjoint means $A \cap B = \varnothing$.
- **Difference:** $A \setminus B = \{x : x \in A \text{ and } x \notin B\} = A \cap B'$ — "in A, but not in B".
- **Complement:** $A' = U \setminus A = \{x \in U : x \notin A\}$. Without U there is no complement — always fix U first.
- **Symmetric difference:** $A \Delta B = (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$ — elements in **exactly one** of the two.

Shaded: A Δ B, the elements belonging to exactly one set



All sets are understood inside a fixed universe U

The algebraic laws (all provable by element-chasing)

KEY RESULT

Commutative: $A \cup B = B \cup A$, $A \cap B = B \cap A$, $A \Delta B = B \Delta A$. Associative: $(A \cup B) \cup C = A \cup (B \cup C)$ (same for $\cap$, Δ). Distributive: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$. Idempotent: $A \cup A = A = A \cap A$. Complement: $A \cup A' = U$, $A \cap A' = \varnothing$, $(A')' = A$. Absorption: $A \cup (A \cap B) = A$, $A \cap (A \cup B) = A$. Difference: $A \setminus B = A \cap B'$, $A \setminus A = \varnothing$.

SHORT DERIVATION

De Morgan's laws

$(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$. Chase elements: $x \in (A \cup B)'$ means "$x \notin A \cup B$", i.e. "$x \notin A$ **and** $x \notin B$" (negation flips "or" to "and"), i.e. $x \in A'$ and $x \in B'$, i.e. $x \in A' \cap B'$. Swapping $\cup$, $\cap$ and negating again gives the second law. The same argument with n sets gives: complement of a union is the intersection of complements, and vice versa — "keep the elements outside **all** of them" $= \bigcap A_i'$, "outside at least one" $= \bigcup A_i'$.

INTUITION

Read the operations as filters on a Venn diagram. $\cup$ = at least one condition true. $\cap$ = all conditions true. $\setminus$ = first true, second false. Δ = odd number of memberships. Every identity becomes "the shaded region is the same picture on both sides" — a genuine proof method once you say which region each element lands in.

Counting with a Venn diagram

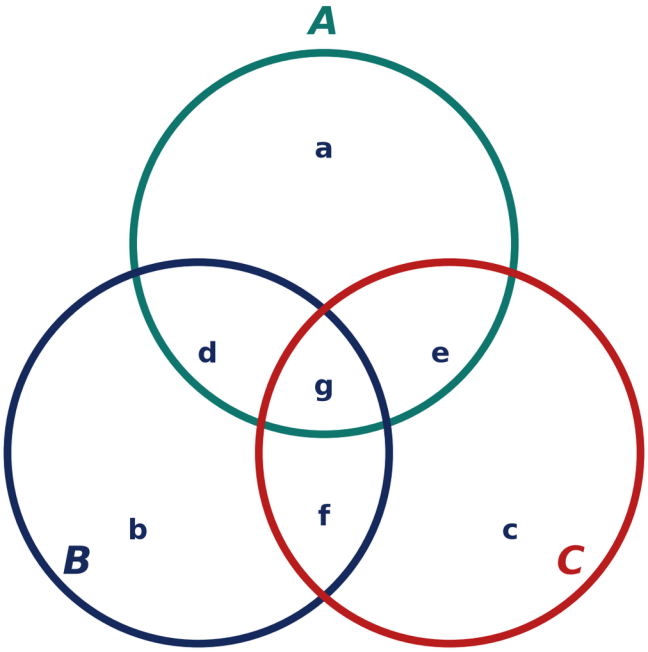
SHORT DERIVATION

Inclusion–exclusion

Partition $A \cup B$ into the three disjoint pieces $A \setminus B$, $A \cap B$, $B \setminus A$. Then $n(A \cup B) = n(A) + n(B) - n(A \cap B)$, because adding $n(A) + n(B)$ counts the overlap twice. For three sets (same idea, seven disjoint regions):

$$n(A \cup B \cup C) = \sum n(A) - \sum n(A \cap B) + n(A \cap B \cap C)$$

where $\sum n(A)$ sums the three singletons and $\sum n(A \cap B)$ the three pairwise intersections.



Seven disjoint regions; n(A∪B∪C) = sum of all seven counts

WORKED EXAMPLE

Standard application. *Of 60 students, 25 play cricket, 20 football, 15 both. How many play neither?* $n(C \cup F) = 25 + 20 - 15 = 30$, *so* $n((C \cup F)') = 60 - 30 = 30$ *students play neither.* ***Check:*** *the 15 who play both are inside both counts; subtracting once leaves each student counted exactly once. Boundary habit: "neither" is a complement, so the total 60 must be the universe.*

COMMON TRAP

- *A complement is meaningless without a stated universe — "the set of students not playing cricket" lives inside the class, not inside all humans.*
- $A \setminus B = \varnothing$ *means* $A \subseteq B$, ***not*** $A = B$.
- $(A \setminus B) \cup B = A \cup B$ *in general, not* A. *The reverse difference* $B \setminus A$ *is a different animal.*
- Δ *is not associative with* $\cup$ *mixing: parentheses matter when three operations combine.*
- $n(A \cup B) = n(A) + n(B)$ *holds* ***only*** *when* $A \cap B = \varnothing$. *Using it without checking disjointness is the classic counting error.*

LESSON 03 / 09

Cartesian product of sets

An **ordered pair** (a, b) remembers order: $(a, b) = (c, d) \Leftrightarrow a = c$ and $b = d$ (this is the working convention; the formal set-theoretic build $(a, b) := \{\{a\}, \{a, b\}\}$ is a background fact, not an exam tool).

KEY RESULT

$A \times B = \{(a, b) : a \in A,\ b \in B\}$. Then $n(A \times B) = n(A) \cdot n(B)$ — product rule: choose the first coordinate ($n(A)$ ways) then the second ($n(B)$ ways), independently.
$A \times B = \emptyset \Leftrightarrow A = \emptyset$ or $B = \emptyset$. $A \times B \neq B \times A$ in general (they are equal only in trivial cases such as $A = B$ or one empty).

Distributive-style facts (all set identities carry over with $\times$):

- $A \times (B \cup C) = (A \times B) \cup (A \times C)$ and $A \times (B \cap C) = (A \times B) \cap (A \times C)$.
- $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$ — intersect **coordinate-wise**.
- $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$.
- $A \times B \subseteq C \times D \Leftrightarrow (A \subseteq C$ and $B \subseteq D)$ when all four are non-empty.

COMMON TRAP

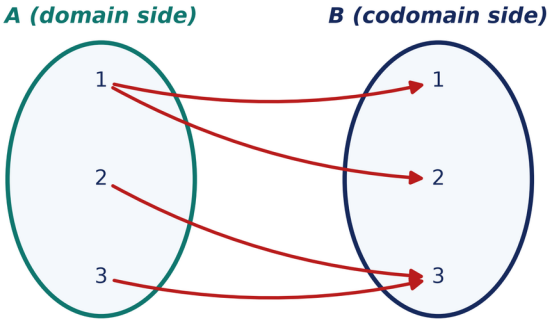
- $(A \times B) \times C \neq A \times (B \times C)$ as sets: the elements are $((a, b), c)$ vs $(a, (b, c))$. Only the elements are in natural bijection. JEE counts pairs, so keep parentheses as written.
- $A \times B$ is a **set of pairs**, not a number: $|A \times B|$ is the number; $A \times B$ itself contains points such as $(1, 4)$.
- $A \times A$ is the plane grid of a set with itself; $n(A \times A) = n(A)^2$, which is why an n-element set has n^2 ordered pairs — memorise this number, you will use it constantly.

Relations

A **relation** R from A to B is any subset $R \subseteq A \times B$. If $B = A$ we say "R is a relation on A". For $(a, b) \in R$ we often write $a\,R\,b$.

- **Domain** $= \{a \in A : \exists b,\ (a, b) \in R\}$ (first coordinates that actually appear).
- **Range** $= \{b \in B : \exists a,\ (a, b) \in R\}$.
- **Codomain** $= B$, fixed by declaration — range is a subset of the codomain, not necessarily equal to it.

Representations: roster form $R = \{(1, 1), (1, 2), (2, 3), (3, 3)\}$; rule form $\{(x, y) : y = x^2\}$; arrow diagrams; the 0-1 matrix ($m_{ij} = 1$ iff $(a_i, b_j) \in R$); graphs for relations on $\mathbb{R}$.



R = {(1,1),(1,2),(2,3),(3,3)}: a general relation — 1 sends two arrows, so R is not a function

Three properties of relations on a set A

- **Reflexive:** $a\,R\,a$ for **every** $a \in A$. One missing diagonal pair kills it.
- **Symmetric:** $a\,R\,b \Rightarrow b\,R\,a$ for all a, b.
- **Transitive:** $a\,R\,b$ and $b\,R\,c \Rightarrow a\,R\,c$.

Standard test cases on $\mathbb{Z}^+$ or $\mathbb{R}$: "$\leq$" is reflexive + transitive, **not** symmetric. "$\neq$" is symmetric, **not** reflexive, **not** transitive ($1 \neq 2$, $2 \neq 1$ but $1 \neq 1$ is false — the transitive conclusion fails). "divides" on $\mathbb{Z}^+$: reflexive + transitive, not symmetric ($2 \mid 4$ but $4 \nmid 2$). Parallelism of lines in a plane (with every line parallel to itself) is an equivalence relation.

SHORT DERIVATION

How many relations exist?

A relation on A is any subset of $A \times A$, and $n(A \times A) = n^2$ for $n = n(A)$. Each of the n^2 pairs is independently in or out of R: total relations $= 2^{n^2}$. Reflexive relations must contain all n diagonal pairs (a, a) and are free on the other $n^2 - n$ pairs: 2^{n^2-n}. Small case $n = 2$, $A = \{1, 2\}$: 4 pairs, $2^4 = 16$ relations; reflexive ones must contain $(1, 1), (2, 2)$ and are choices on $(1, 2), (2, 1)$: $2^2 = 4$ — list them: $\{(1, 1), (2, 2)\}$, plus each of $(1, 2)$ and/or $(2, 1)$.

WORKED EXAMPLE

Worked. $A = \{1, 2, 3\}$, $B = \{4, 5\}$. Then $A \times B$ has 6 pairs, so there are $2^6 = 64$ relations from A to B. Let $R = \{(x, y) \in A \times B : y - x \geq 3\}$. Check all six pairs: $(1, 4) : 4 - 1 = 3 \geq 3$ ✓; $(1, 5) : 4 \geq 3$ ✓; $(2, 4) : 2 \geq 3$ ✗; $(2, 5) : 3 \geq 3$ ✓; $(3, 4) : 1 \geq 3$ ✗; $(3, 5) : 2 \geq 3$ ✗. So $R = \{(1, 4), (1, 5), (2, 5)\}$, domain $= \{1, 2\}$, range $= \{4, 5\}$, codomain $= \{4, 5\}$. **Check:** the boundary $y - x = 3$ is included ($\geq$, not $>$), which is what makes $(1, 4)$ and $(2, 5)$ members — boundary conditions decide membership more often than bulk algebra.

LESSON 04 / 09

Equivalence relation

A relation R on A is an **equivalence relation** if it is reflexive, symmetric **and** transitive. The model example is congruence: for fixed $n \geq 2$, define $x\,R\,y \Leftrightarrow n \mid (x-y)$ ("$x \equiv y \pmod n$").

SHORT DERIVATION

Congruence mod n is an equivalence relation on Z

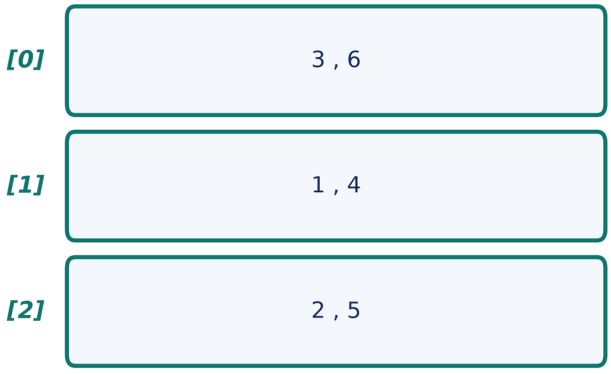
Reflexive: $x - x = 0$ and $n \mid 0$. Symmetric: if $n \mid (x-y)$ then $x - y = nk$, so $y - x = n(-k)$ and $n \mid (y-x)$. Transitive: $n \mid (x-y)$ and $n \mid (y-z)$ give $x - y = nk$, $y - z = nl$, so $x - z = n(k+l)$. Each step is one divisibility fact — nothing clever, which is exactly why this template reappears in exams.

Equivalence classes and partitions

For $a \in A$ the **class** $[a] = \{x \in A : x\,R\,a\}$. Fundamental facts: $a \in [a]$ (reflexivity); $[a] = [b] \Leftrightarrow a\,R\,b$; distinct classes are disjoint; the classes cover A. So the classes form a **partition** of A: a family of non-empty, pairwise disjoint blocks whose union is A.

KEY RESULT

Equivalence relations on A and partitions of A are the same data in two languages. $a\,R\,b \Leftrightarrow$ "a and b are in the same block". Counting either one gives the **Bell numbers**: $B_1 = 1$, $B_2 = 2$, $B_3 = 5$, $B_4 = 15$, $B_5 = 52$ — the number of equivalence relations on an n-element set. For $A = \{1,2,3\}$ the five partitions are: $\{\{1,2,3\}\}$; $\{\{1\},\{2,3\}\}$, $\{\{2\},\{1,3\}\}$, $\{\{3\},\{1,2\}\}$; $\{\{1\},\{2\},\{3\}\}$.



Partition induced by congruence mod 3 on {1,...,6}: three classes, one block each

WORKED EXAMPLE

Worked. On $\mathbb{Z}$ let $x\,R\,y \Leftrightarrow x+y$ is even. Symmetric is immediate ($x+y=y+x$); reflexive: $x+x=2x$ even; transitive: if $x+y$ and $y+z$ are even, then $(x+y)-(y+z)=x-z$ is even and $x+z=(x-z)+2z$ is even. Classes: even numbers $[0]=\{\dots,-2,0,2,\dots\}$ and odds $[1]=\{\dots,-1,1,3,\dots\}$. **Check:** $[2]=[0]$ since $2+0$ is even — classes are named by any member, not only by the smallest. Note this R is the same equivalence as $x\equiv y \pmod 2$.

The traps that generate exam questions

COMMON TRAP

1. ***Symmetric + transitive does not imply reflexive.*** On $A=\{1,2,3\}$ the relation $R=\{(1,1),(1,2),(2,1),(2,2)\}$ is symmetric and transitive but not reflexive ($3\notin$ any pair with itself).
2. ***The empty relation on a non-empty*** A is symmetric and transitive (vacuously) but not reflexive. The ***universal relation*** $A\times A$ is an equivalence relation (one big block). The ***identity relation*** $\{(a,a):a\in A\}$ is also one (all singleton blocks).
3. ***To prove "equivalence"*** check the three axioms separately on arbitrary elements; a single counterexample kills any axiom. Many partial proofs assume transitivity "by symmetry" — never valid.
4. ***Class names:*** $[1]=[4]$ under mod 3; the partition blocks are sets of elements, the labels are cosmetic.

How JEE counts these (derivable with the syllabus)

SHORT DERIVATION

Reflexive and symmetric relations on an n-set

Reflexive + symmetric: the n diagonal pairs are forced in; off-diagonal, (a,b) and (b,a) must be chosen together — in or out as a pair. There are $\frac{n(n-1)}{2}$ such unordered pairs, so the count is $2^{n(n-1)/2}$. Small case $n=3$: $2^3=8$ relations — forced $\{(1,1),(2,2),(3,3)\}$ plus any subset of $\{(1,2),(2,1)\},\{(1,3),(3,1)\},\{(2,3),(3,2)\}$, listed as $2^3=8$ choices. Similarly: reflexive alone $=2^{n^2-n}$, symmetric alone $=2^{n(n+1)/2}$ (diagonal free + pairs together), antisymmetric $=2^n\cdot 3^{n(n-1)/2}$ (diagonal free; each off-diagonal pair: neither, only (a,b), or only (b,a)).

STANDARD APPLICATION

Standard application. *Number of relations on $\{1,2,3\}$ that are reflexive and symmetric $= 2^3 = 8$; that are equivalence relations $= B_3 = 5$; that are reflexive $= 2^6 = 64$. The examiner tests whether you separate the three axioms' constraints — diagonal forced, pairs coupled, or fully free.*

LESSON 05 / 09

A function is a relation with two guarantees

$f: A \to B$ is a relation $f \subseteq A \times B$ such that every $a \in A$ has **exactly one** $b \in B$ with $(a, b) \in f$. The two halves — existence (at least one image) and uniqueness (at most one) — are checked separately on exam questions. A is the **domain** (declared; it is part of the definition), B the **codomain**, and $f(A) = \{f(a) : a \in A\} \subseteq B$ the **range** (image).

- Arrow-diagram test: every element of A sends exactly one arrow out of B (an element of B may receive many or none).
- Graph test on $\mathbb{R}$: vertical line test (every vertical line hits the curve at most once) plus domain coverage.

Notation: $y = f(x)$; $f: x \mapsto x^2 - 1$; f and g are equal ($f = g$) iff same domain, same codomain and same values everywhere.

ASSUMPTIONS & CONVENTIONS

Statements like "find the domain of f" assume the **maximal** real domain unless a domain is specified: all real x for which the formula makes sense. Range is always inside the declared codomain; when a question says $f: \mathbb{R} \to \mathbb{R}$ and asks whether f is onto, the codomain being all of $\mathbb{R}$ is part of the data. Writing $f: A \to B$ with A the natural domain and B chosen larger than the range makes f "into" but not onto — same formula, different function.

Finding domains (the rule stack)

KEY RESULT

Apply every restriction simultaneously, then intersect:

1. Polynomial: all of $\mathbb{R}$. 2. Rational $p(x)/q(x)$: $q(x) \neq 0$. 3. Even root $\sqrt[n]{g(x)}$ (n even): $g(x) \geq 0$. 4. Logarithm $\log_b g(x)$: $g(x) > 0$, base $b > 0, b \neq 1$. 5. $\tan g(x)$: $g(x) \neq \frac{\pi}{2} + k\pi$. 6. $1/g(x)$: additionally $g(x) \neq 0$. 7. Composite $g \circ f$: $x \in \operatorname{dom} f$ **and** $f(x) \in \operatorname{dom} g$. 8. Real powers $g^{p/q}$ in lowest terms: if q even, need $g \geq 0$.

WORKED EXAMPLE

Worked (domain). $f(x) = \sqrt{x - 1} + \dfrac{\sqrt{5 - x}}{x - 3}$. Conditions: $x - 1 \geq 0 \Rightarrow x \geq 1$; $5 - x \geq 0 \Rightarrow x \leq 5$; $x - 3 \neq 0 \Rightarrow x \neq 3$. Domain $= [1, 3) \cup (3, 5]$. **Boundary checks:** $x = 1$: $f(1) = 0 + \frac{2}{-2} = -1$ — defined, so the bracket at 1 is closed. $x = 5$: $f(5) = 2 + 0$ — defined, closed. $x = 3$: division by zero — deleted even though both roots are fine there. Endpoint testing is not optional: open vs closed brackets are where partial marks live.

Finding ranges (the four techniques)

- **Complete the square:** $x^2 - 4x + 7 = (x-2)^2 + 3 \geq 3$, so range on $\mathbb{R}$ is $[3, \infty)$; the vertex $(2, 3)$ is the only point where the bound is met.
- **Monotone pieces:** a strictly decreasing f maps $[a, b]$ onto $[f(b), f(a)]$ (order flips!). On $[0, 1]$, $x^2 - 4x + 7$ decreases from 7 to 4: range $[4, 7]$.
- **Solve for x:** write $y = f(x)$, solve for $x = g(y)$, and require a real solution: for $y = \dfrac{x^2+1}{x^2-1}$ with $x^2 \neq 1$... cleaner classic below.
- **Sign / bound analysis:** $|f|$, $e^x > 0$, $\sqrt{\cdot} \geq 0$ give one-sided bounds; combine with continuity (the image of an interval is an interval).

WORKED EXAMPLE

Worked (range by discriminant). $f(x) = \dfrac{x^2}{x^2+1}$ on $\mathbb{R}$. Solve $y = \frac{x^2}{x^2+1}$ for x: $y(x^2+1) = x^2 \Rightarrow x^2(1-y) = y \Rightarrow x^2 = \frac{y}{1-y}$. Real x exists iff $\frac{y}{1-y} \geq 0$, i.e. $0 \leq y < 1$ (note $y = 1$ is impossible: $1 - y = 0$ gives $0 = 1$). Range $= [0, 1)$. **Check:** $f(0) = 0$ attained; $f(x) \to 1$ as $|x| \to \infty$ but never equals 1 — the range is half-open exactly as the algebra says.

COMMON TRAP

- $\sqrt{x^2} = |x|$, not x — the root output is never negative.
- $e^{\ln x} = x$ only for $x > 0$; $\ln(e^x) = x$ for all real x. Swapping them silently changes domains.
- Range of a formula **changes with the domain**: x^2 on $\mathbb{R}$ has range $[0, \infty)$; on $[-1, 2]$ it has range $[0, 4]$.
- $f(A) \cap f(B) \supseteq f(A \cap B)$ always, but equality can fail (e.g. $f(x) = x^2$, $A = \{-2\}$, $B = \{2\}$): $f(A \cap B) = \varnothing$ while $f(A) \cap f(B) = \{4\}$. The similar-looking $f(A \setminus B)$ vs $f(A) \setminus f(B)$ is even more fragile — draw arrows before trusting algebra.

Image and preimage (defined for every function)

For $S \subseteq A$ the **image** is $f(S) = \{f(x) : x \in S\}$. For $T \subseteq B$ the **preimage** is $f^{-1}(T) = \{x \in A : f(x) \in T\}$. The superscript -1 here does **not** require invertibility — $f^{-1}(T)$ is a set of inputs, and it can be empty ($f^{-1}(\{-5\}) = \varnothing$ for $f(x) = x^2$).

KEY RESULT

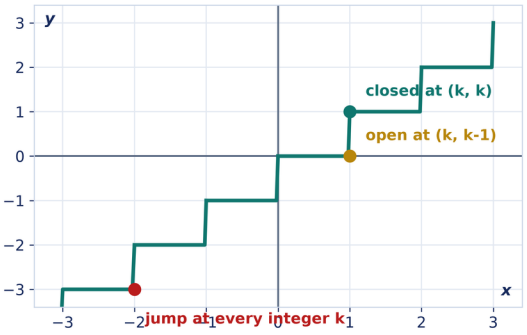
f^{-1} used as a **function** exists only for bijections (see the composition & inverse sheet). $f^{-1}(T)$ used as a **set** is always defined. Identities that always hold: $f(f^{-1}(T)) \subseteq T$; $f^{-1}(f(S)) \supseteq S$; $f^{-1}(T_1 \cap T_2) = f^{-1}(T_1) \cap f^{-1}(T_2)$ (both directions); $f(S_1 \cup S_2) = f(S_1) \cup f(S_2)$.

LESSON 06 / 09

The standard library (know the graphs cold)

- **Absolute value:** $|x| = x$ for $x \geq 0$, $-x$ for $x < 0$. Always ≥ 0; $|x| = \sqrt{x^2}$; triangle inequality $|a+b| \leq |a| + |b|$ with equality iff a, b have the same sign (or one is 0).
- **Signum:** $\operatorname{sgn}(x) = 1$ $(x > 0)$, 0 $(x = 0)$, -1 $(x < 0)$. So $x = |x|\operatorname{sgn}(x)$ for all x.
- **Greatest integer (floor):** $[x] = \lfloor x \rfloor$ = greatest integer $\leq x$. Fundamental sandwich: $[x] \leq x < [x] + 1$, equivalently $x = [x] + \{x\}$ with **fractional part** $\{x\} = x - [x] \in [0, 1)$.
- **Exponential / logarithm:** $e^x : \mathbb{R} \to (0, \infty)$ strictly increasing; $\ln x : (0, \infty) \to \mathbb{R}$ its inverse. $e^{\ln x} = x$ $(x > 0)$, $\ln e^x = x$ (all x).

Floor function y = [x]: right-continuous jumps at every integer (open circle = value excluded, closed = included)



COMMON TRAP

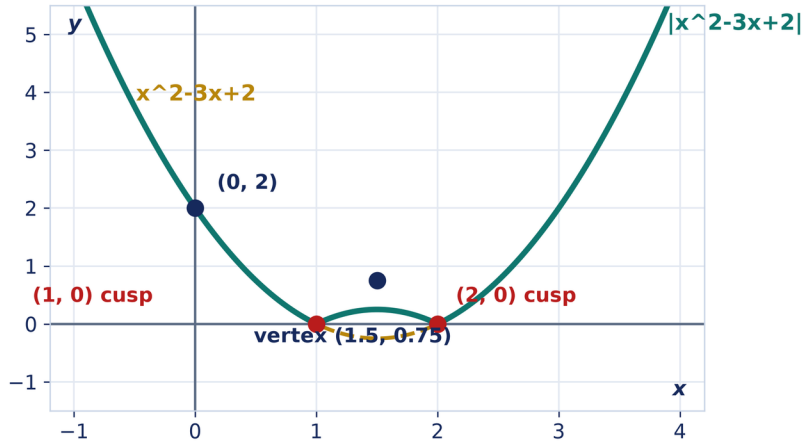
- $\sqrt{x^2} = |x|$, never x: $\sqrt{(-3)^2} = 3$, not -3.
- $[x]$ jumps by 1 at integers: $[0.999] = 0$ but $[1] = 1$. Draw open/closed dots explicitly — $[x]$ is **right-continuous**: value at the jump equals the right limit.
- $[x] + [-x] = 0$ if $x \in \mathbb{Z}$, else -1. $\{x\} + \{-x\} = 0$ if $x \in \mathbb{Z}$, else 1.
- $e^x > 0$ always: "$e^x = -2$" has no solution. $\ln x$ of a negative argument is undefined in $\mathbb{R}$.
- $|x| = x$ iff $x \geq 0$ (equality includes 0 — boundary!).

Graph transformations (one picture, many curves)

KEY RESULT

From $y = f(x)$: $f(x) + c$ shifts **up** c; $f(x + c)$ shifts **left** c (mind the sign); $af(x)$ scales vertically; $f(ax)$ scales horizontally by $1/|a|$ and reflects if $a < 0$; $|f(x)|$ reflects every negative lobe **upward** (creates cusps at old zeros); $f(|x|)$ keeps $x \geq 0$ and mirrors it into $x < 0$ (even function); $\{f(x)\}$ wraps each value into $[0, 1)$ (sawtooth along the curve); $1/f(x)$ turns zeros into vertical asymptotes and bounds into horizontal ones.

y = |x^2 - 3x + 2| (solid) from y = x^2 - 3x + 2 (dashed): roots at 1, 2 become cusps



WORKED EXAMPLE

Worked. Sketch $y = \{x\} = x - [x]$: straight segments of slope 1 from each integer up to, but not including, the next integer height 1; value 0 exactly at integers (closed dots at $(k, 0)$, open dots at $(k, 1)$). Range $[0, 1)$ — half-open, every time. Then $\{3.7\} = 0.7$, $\{-1.2\} = 0.8$ (because $[-1.2] = -2$). **Check:** $\{x\} \in [0, 1)$ for negatives too — the floor of a negative number is the integer to its **left**, not "the integer part" of the digits.

STANDARD APPLICATION

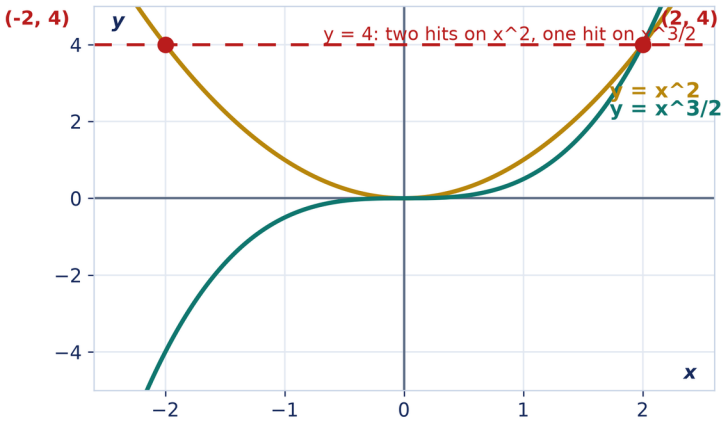
Standard application. Solve $|x - 1| + |x - 3| = 4$ by breaking at the knots $x = 1, 3$: on $x < 1$: $(1 - x) + (3 - x) = 4 \Rightarrow x = 0$ ✓; on $1 \le x < 3$: $(x - 1) + (3 - x) = 2 \neq 4$; on $x \ge 3$: $(x - 1) + (x - 3) = 4 \Rightarrow x = 4$ ✓. Solution set $\{0, 4\}$. **Check boundaries:** $x = 1$ gives $0 + 2 = 2$; $x = 3$ gives $2 + 0 = 2$ — knots must be tested separately, they belong to exactly one case each.

LESSON 07 / 09

Injective, surjective, bijective

- **One-one (injective):** $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. Test: horizontal line test on the graph; or solve $f(x_1) = f(x_2)$ and cancel. Strictly monotone (increasing or decreasing) on an interval $\Rightarrow$ one-one there.
- **Onto (surjective):** range = codomain. "Into" just means not onto. To prove onto: for arbitrary y in the codomain, exhibit x with $f(x) = y$ and check x lies in the domain.
- **Bijection:** both. Exactly then does the inverse function f^{-1} exist (see the composition & inverse sheet).

Horizontal line test: y = x^2 is not one-one on R (line y = 4 crosses twice) but y = x^3 is



SHORT DERIVATION

How many functions A to B?

Each of the $n(A)$ inputs independently chooses one of the $n(B)$ outputs — product rule: $|B|^{|A|}$ functions. Among them: **injections** $= n(B)(n(B)-1)\cdots(n(B)-n(A)+1) = P(n(B), n(A))$ if $n(A) \le n(B)$, else 0 (pigeonhole: more inputs than outputs forces a collision). **Bijections** $= n(A)!$ when $n(A) = n(B)$. **Surjections:** if $n(A) < n(B)$ none exist; otherwise count by cases for tiny sets — $f: \{1,2,3\} \to \{a,b\}$ has $2^3 = 8$ functions total, exactly 2 constant, so $8 - 2 = 6$ onto (each value must appear, i.e. not all three go to one side). The general inclusion–exclusion formula is Part II material; this case analysis is fully syllabus-level.

KEY RESULT

Small-case table for $|A| = 3, |B| = 2$: total functions $= 8$; one-one $= 0$; onto $= 6$; constant $= 2$. For $|A| = 2, |B| = 3$: total $= 9$; one-one $= 3 \cdot 2 = 6$; onto $= 0$; constant $= 3$. Memorise the shape of the table, not just the numbers — the constraints ($n(A) \le n(B)$ for injective, $n(A) \ge n(B)$ for surjective) are what get tested.

Even and odd functions

Requires a domain **symmetric about 0** (if x is allowed so is $-x$). Then f is **even** if $f(-x) = f(x)$ (graph symmetric about the y-axis), **odd** if $f(-x) = -f(x)$ (graph symmetric about the origin; rotation by 180°).

SHORT DERIVATION

Every function splits into even + odd parts

$f(x) = g(x) + h(x)$ where $g(x) = \frac{f(x)+f(-x)}{2}$ (even part) and $h(x) = \frac{f(x)-f(-x)}{2}$ (odd part). Check: $g(-x) = \frac{f(-x)+f(x)}{2} = g(x)$; $h(-x) = \frac{f(-x)-f(x)}{2} = -h(x)$. The split is unique. Consequences worth knowing: an odd f with 0 in its domain has $f(0) = 0$; even $\pm$ even = even, odd $\pm$ odd = odd; even $\times$ odd = odd; $f \cdot f$ is even whenever f is odd or even.

COMMON TRAP

- $f(x) = x^2$ on $[0, 1]$ is **neither** even nor odd — the domain is not symmetric. Check the domain before the formula.
- $f(x) = x + 1$ is neither (sum of odd x and even 1 — sums of mixed parity are usually neither).
- $f(0) = 0$ is necessary for odd functions defined at 0, not sufficient ($f(x) = x^3 + x$ odd ✓, $f(x) = x^2$ has $f(0) = 0$ but is even).

Periodic functions

f is **periodic** with period $T > 0$ if $f(x + T) = f(x)$ for all x (domain must allow $x + T$). The least such T is the **fundamental period** ("the period" in exam language).

KEY RESULT

Periods: $\sin x, \cos x$: 2π; $\tan x$: π; $\sin(ax + b), \cos(ax + b)$: $\frac{2\pi}{|a|}$; $\tan(ax + b)$: $\frac{\pi}{|a|}$; $|\sin x|$, $\sin^2 x$, $\cos^2 x$: π (taking absolute value or squaring merges the two halves); $\lfloor x \rfloor$, $\{x\}$: 1; a constant function satisfies $f(x + T) = f(x)$ for every T — it is periodic but has **no** fundamental period.

WORKED EXAMPLE

Worked. Period of $f(x) = \sin\left(\frac{x}{2}\right) + \cos(3x)$. First term: $\frac{2\pi}{1/2} = 4\pi$. Second: $\frac{2\pi}{3}$. Candidate period $= \operatorname{lcm}(4\pi, \frac{2\pi}{3}) = 4\pi$ (need $T = k \cdot 4\pi = m \cdot \frac{2\pi}{3}$; least positive is 4π). **Check:** $f(x + 4\pi) = \sin(\frac{x}{2} + 2\pi) + \cos(3x + 12\pi) = f(x)$ ✓. Trap: lcms only exist for commensurable periods ($T_1/T_2 \in \mathbb{Q}$); $\sin x + \sin(\pi x)$ is periodic only in trivial senses — never assume a combined period exists.

STANDARD APPLICATION

Standard application. $f(x) = \sin^2\left(\frac{x}{2}\right)$: use $\sin^2\theta = \frac{1-\cos 2\theta}{2} = \frac{1-\cos x}{2}$, whose period is 2π — while naive reading suggests $\frac{2\pi}{1/2} = 4\pi$ for $\sin(x/2)$ then squaring halves it to 2π. Reduce with identities first, then read off the period; the two-stage guess 4π is wrong.

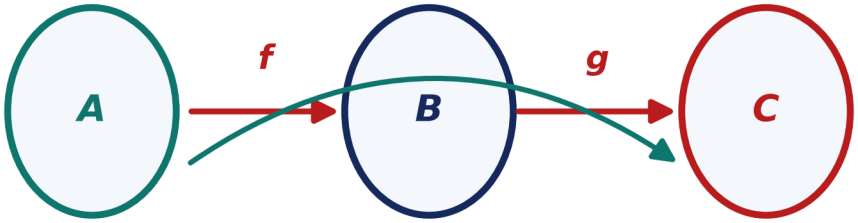
LESSON 08 / 09

Composition: $g \circ f$

$(g \circ f)(x) = g(f(x))$ — apply f **first**, then g.

KEY RESULT

Domain of $g \circ f$: $\{x \in \operatorname{dom} f : f(x) \in \operatorname{dom} g\}$. Order matters: $g \circ f \neq f \circ g$ in general. Composition is associative ($(h \circ g) \circ f = h \circ (g \circ f)$) because both sides equal $h(g(f(x)))$. If $f : A \to B$ and $g : B \to C$ then $g \circ f : A \to C$ and $f(A) \subseteq B$ must land inside g's domain — if f's range escapes $\operatorname{dom} g$, shrink f's domain until it does not.



g o f : A to C

WORKED EXAMPLE

Worked. $f(x) = \sqrt{x}$ (domain $[0, \infty)$), $g(x) = x - 3$ (domain $\mathbb{R}$). Then $(g \circ f)(x) = \sqrt{x} - 3$ with domain $[0, \infty)$ — every output of f is fine for g. But $(f \circ g)(x) = \sqrt{x - 3}$ has domain $[3, \infty)$ — g must not leave f's domain. **Check:** $x = 0$: $g(f(0)) = -3$ defined; $f(g(0)) = \sqrt{-3}$ undefined. The same two functions give different composite domains; the direction of composition is the whole question.

Inverse functions

$f : A \to B$ has an **inverse** $f^{-1} : B \to A$ iff f is bijective; then $f^{-1}(f(x)) = x$ for $x \in A$ and $f(f^{-1}(y)) = y$ for $y \in B$. Recipe: write $y = f(x)$, solve for $x = g(y)$, swap letters: $f^{-1}(x) = g(x)$. The domain of f^{-1} is the **range** of f and vice versa.

SHORT DERIVATION

(g o f) inverse rule

$(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ — the socks-and-shoes law: undo g first, then f. Check: $(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ \mathrm{id} \circ f = \mathrm{id}$ ✓, and symmetrically on the other side (needed since $g \circ f$ bijective implies each of f, g suitably bijective between the right sets).

Graphically $y = f^{-1}(x)$ is the reflection of $y = f(x)$ in the line $y = x$ (points $(a, b) \mapsto (b, a)$). If f is one-one but not onto, restrict the codomain to the range; if onto but not one-one, restrict the domain (e.g. x^2 on $[0, \infty)$ has inverse $\sqrt{x}$).

WORKED EXAMPLE

Worked. $f(x)=\frac{2x+3}{x-1}$, $x\neq 1$. Solve $y=\frac{2x+3}{x-1}$: $y(x-1)=2x+3\Rightarrow x(y-2)=y+3\Rightarrow x=\frac{y+3}{y-2}$. So $f^{-1}(x)=\frac{x+3}{x-2}$, domain $x\neq 2$ = range of f. Vertical asymptote $x=1$, horizontal asymptote $y=2$ of f become the horizontal $y=1$ and vertical $x=2$ of f^{-1} — asymptotes swap roles under reflection in $y=x$, exactly like domain and range. **Check:** $f(f^{-1}(3))=f(6)=\frac{15}{5}=3$ ✓; $f^{-1}(f(2))=f^{-1}(7)=\frac{10}{5}=2$ ✓.

COMMON TRAP

- f^{-1} as an inverse **function** needs bijection; writing f^{-1} before checking one-one + onto is the commonest Chapter-1 error.
- $(f\circ g)^{-1}\neq f^{-1}\circ g^{-1}$ (order flips!). The square bracket notation $f^{-1}(x)\neq\frac{1}{f(x)}$ — the -1 superscript means inverse function (or preimage set), never reciprocal, except the (rare) $\sin^{-1}$ convention confusion: $\sin^{-1}x$ is the inverse sine, while $(\sin x)^{-1}=\csc x$.
- $f\circ g=g\circ f$ happens only in special cases (e.g. $f=g$; or both powers x^a, x^b; or $f=g^{-1}$ giving identity) — never assume it.

Chapter recap — the whole sheet in one place

KEY RESULT

Sets: $|\mathcal{P}(A)|=2^n$; equal = double inclusion; $\varnothing$, $\{0\}$, $\{\varnothing\}$ distinct. Operations: De Morgan $(A\cup B)'=A'\cap B'$; $A\setminus B=A\cap B'$; $A\Delta B=(A\cup B)\setminus(A\cap B)$; $n(A\cup B)=n(A)+n(B)-n(A\cap B)$; three-set version with alternating signs. Products & relations: $n(A\times B)=n(A)n(B)$; relations on n-set $=2^{n^2}$; reflexive $=2^{n^2-n}$; symmetric $=2^{n(n+1)/2}$; reflexive+symmetric $=2^{n(n-1)/2}$; equivalence = partitions $=B_n$. Functions: $|B|^{|A|}$ total; injections $P(|B|,|A|)$; bijections $n!$; domain rule stack; $\mathrm{dom}(g\circ f)$ careful; f invertible iff bijective; $(g\circ f)^{-1}=f^{-1}\circ g^{-1}$; even/odd need symmetric domain; periods scale as $1/|a|$.

STANDARD APPLICATION

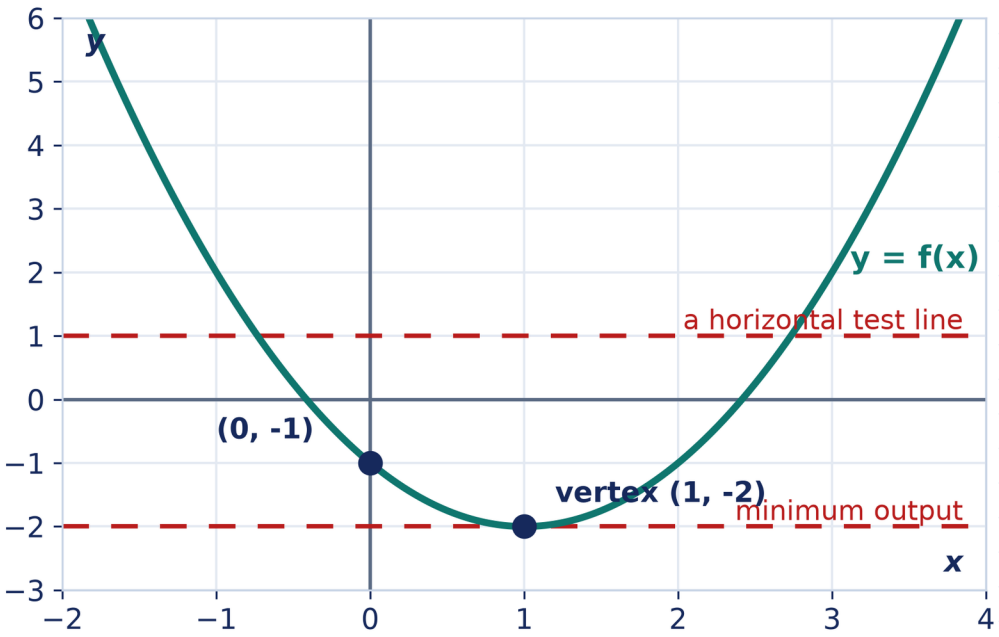
How JEE asks this chapter. (i) Domain/range of built formulas with roots, logs, floors — always test endpoints. (ii) Counting: number of relations/functions with axiom constraints. (iii) Equivalence classes: partition a concrete set, name classes. (iv) Function classification: one-one/onto/even/odd/periodic verdicts on a given formula. (v) Composite domains and inverse formulas with parameter conditions. Each is a two-line application of the boxes above — the marks go to boundary discipline, not to new algebra.

LESSON 09 / 09

Read a graph in three passes

A graph is not decoration: it is the set of ordered pairs $(x, f(x))$. Read it in this order. **First**, project the curve onto the x-axis to obtain the domain. **Second**, project it onto the y-axis to obtain the range. **Third**, use vertical and horizontal lines to ask whether an input is repeated and whether an output has more than one input.

One curve, three kinds of information



READ THE FIGURE
The vertex fixes the minimum output. The horizontal line $y = 1$ meets the curve twice, so this unrestricted quadratic is not one-one.

For $f(x) = x^2 - 2x - 1 = (x-1)^2 - 2$, the domain is $\mathbb{R}$ and the range is $[-2, \infty)$. A vertical line meeting a relation more than once would violate the function condition. A horizontal line meeting a function more than once shows failure of injectivity; these are different tests answering different questions.

COMMON TRAP

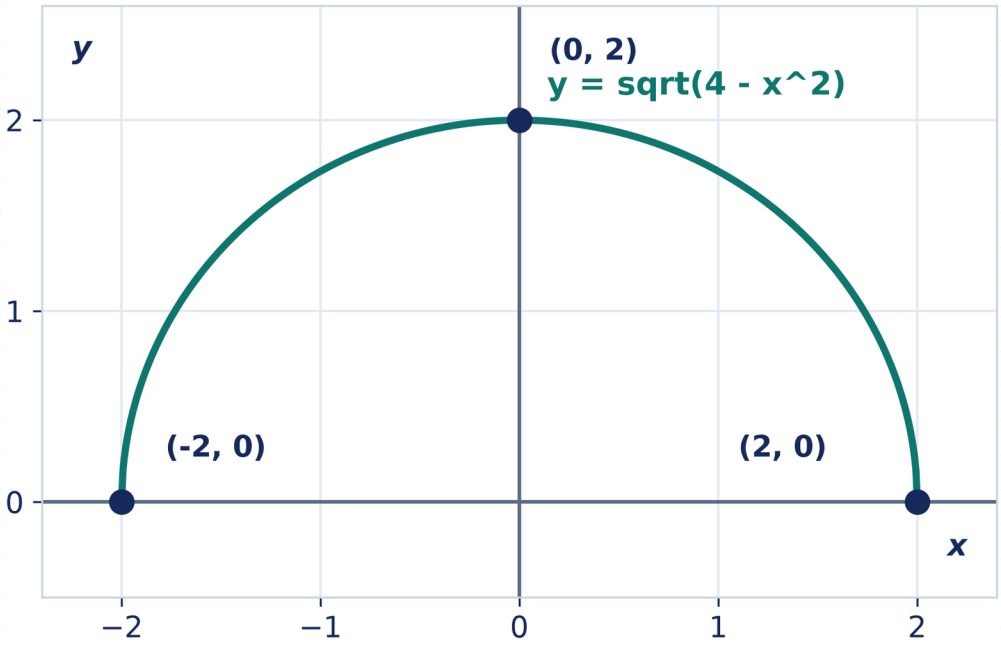
Do not confuse the two line tests

The vertical-line test asks, "Is this relation a function?" The horizontal-line test asks, "Is this function one-one?" Neither test decides onto until a codomain is specified. For example, $x^2 : \mathbb{R} \to \mathbb{R}$ is not onto, but $x^2 : \mathbb{R} \to [0, \infty)$ is onto.

A domain-and-range example drawn before algebra

Consider $g(x)=\sqrt{4-x^2}$. The square root first requires $4-x^2\ge 0$, so $-2\le x\le 2$. On that domain, $0\le 4-x^2\le 4$; taking the nonnegative square root gives $0\le g(x)\le 2$. Both endpoints are attained: $g(\pm 2)=0$ and $g(0)=2$.

The upper semicircle explains both domain and range



READ THE FIGURE
The curve exists only between x = -2 and x = 2. Its height runs continuously from 0 to 2, with every intermediate value attained.

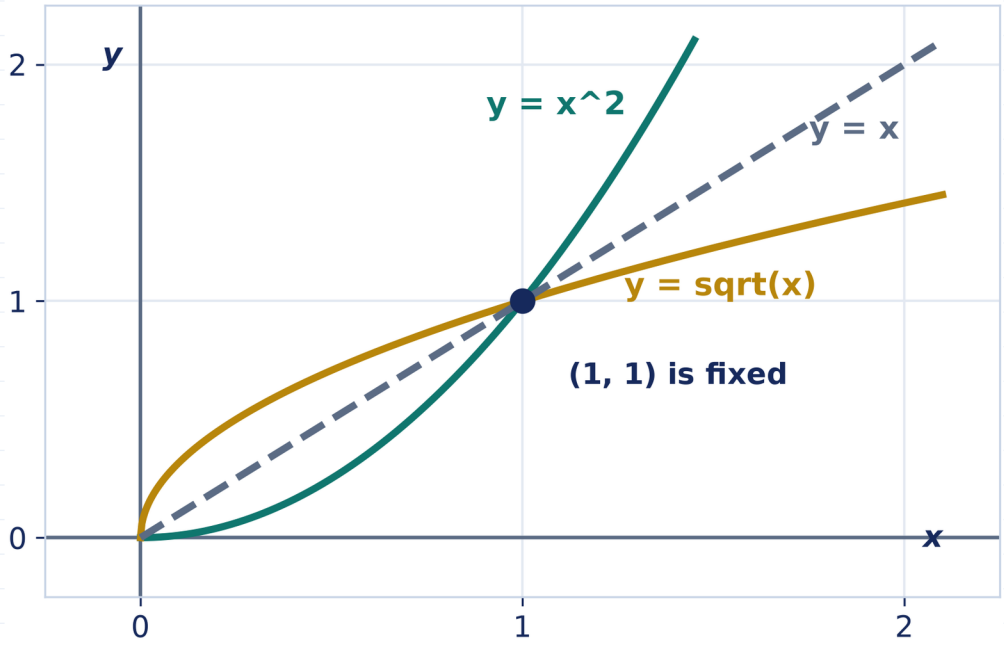
The algebraic calculation and the picture should agree. Squaring $y=\sqrt{4-x^2}$ gives $x^2+y^2=4$, but the original equation also says $y\ge 0$. Therefore the graph is only the **upper** semicircle, not the whole circle. This extra sign condition is exactly the information lost by squaring.

One-one becomes visible after restriction

On $[0,\infty)$, the function $f(x)=x^2$ is strictly increasing and hence one-one. Its inverse is $f^{-1}(x)=\sqrt{x}$. Reflecting the point (a,b) in $y=x$ gives (b,a), so the graphs of a function and its inverse are mirror images in that line.

Inverse graphs interchange the coordinates

Inverse graphs interchange the coordinates



READ THE FIGURE

The restriction x >= 0 is essential. Without it, x^2 is not one-one and no single inverse function exists on the whole real line.

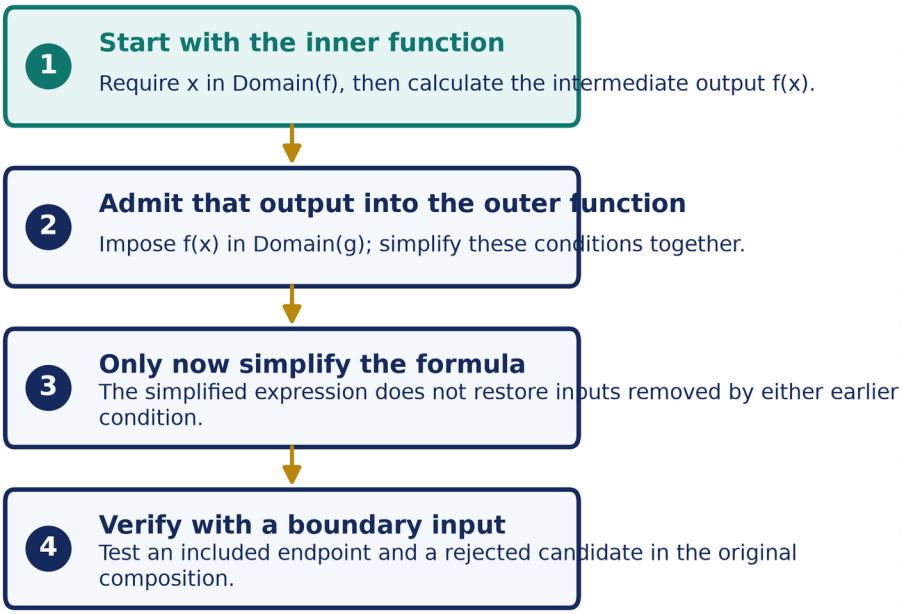
To find an inverse algebraically, write $y = f(x)$, solve for x in terms of y, and then interchange the variable names. Carry the branch restriction through every step. Finally check both $f(f^{-1}(x)) = x$ on the codomain of f and $f^{-1}(f(x)) = x$ on the domain of f.

Composition: check the inner output first

For $(g \circ f)(x) = g(f(x))$, the number x must belong to the domain of f, **and** $f(x)$ must belong to the domain of g. The second condition is easy to forget because the final simplified formula may appear to accept more inputs.

A reusable composition check

Domain conditions travel through the composition



READ THE FIGURE

Domain conditions travel from left to right through the composition; they are not recovered by looking only at the final expression.

WORKED EXAMPLE

Two orders, two different domains

Let $f(x)=\sqrt{x}$ and $g(x)=1-x$. Then $(g\circ f)(x)=1-\sqrt{x}$, with domain $[0,\infty)$. In the other order, $(f\circ g)(x)=\sqrt{1-x}$, with domain $(-\infty,1]$. At $x=4$, the first composition gives -1, while the second is undefined because $g(4)=-3$ cannot enter f. The formulas and domains both show why composition is generally not commutative.

KEY RESULT

The visual revision routine

For any graph, mark the domain on the x-axis and the range on the y-axis. Draw a vertical line for the function test and a horizontal line for the one-one test. For an inverse, draw $y=x$ and check reflection. For a composition, write the intermediate output explicitly. These four drawings prevent most avoidable function errors.

Worked synthesis: composition, range and inversion

Let $h(x)=\sqrt{1-x^2}$ with the declared domain $[-1,1]$. The inner square accepts every real x, but the outer square root requires $1-x^2\ge 0$, which is exactly the declared domain. Keeping both conditions visible prevents an apparently larger domain from appearing after simplification.

WORKED EXAMPLE

Compute the range from the inner output

On $[-1,1]$, the inner quantity $u=x^2$ ranges over $[0,1]$. As u increases, $\sqrt{1-u}$ decreases from 1 to 0. Therefore the range is $[0,1]$, with both endpoints attained: $h(0)=1$ and $h(1)=0$.

WORKED EXAMPLE

Restrict first, then invert

The formula x^2 is not one-one on $[-1,1]$, so the initial squaring step does not have an unrestricted inverse. On $[0,1]$, however, h is strictly decreasing and $h(h(x))=\sqrt{1-(1-x^2)}=\sqrt{x^2}=x$. Thus h is its own inverse on $[0,1]$. The absolute-value step uses $x\ge 0$; it is not optional.

KEY RESULT

One example, three checks

Domain comes from the outer function's requirement, range follows the inner output interval, and inversion requires a one-one restriction. Algebraic symmetry alone supplies none of these three hypotheses.

LESSON 01 / 03

Status of this sheet

Part I derived total, reflexive, symmetric and reflexive+symmetric counts with direct pair analysis. This sheet completes the catalogue (antisymmetric, equivalence/Bell, surjections by inclusion–exclusion) and shows the small-case listings an IOQM/RMO solution must carry. All counts are **models built from the Part I supply list** (product rule, complements, inclusion–exclusion on sets) — the technique is exam-derivable even where the topic is listed as an extension.

The catalogue for $|A| = n$

KEY RESULT

On an n-element set: relations $= 2^{n^2}$; reflexive $= 2^{n^2-n}$; symmetric $= 2^{n(n+1)/2}$; antisymmetric $= 2^n \cdot 3^{n(n-1)/2}$; reflexive+symmetric $= 2^{n(n-1)/2}$; reflexive+transitive = number of "preorder with fixed diagonal" (no simple closed form — casework); equivalence relations $= B_n$ (Bell): $1, 2, 5, 15, 52$ for $n = 1..5$. Functions $A \to B$ with $|B| = m$: m^n; injective: $\frac{m!}{(m-n)!}$ if $n \le m$ else 0; surjective: $n!\,S(n,m)$ with S the Stirling number $= \sum_{i=0}^{m} (-1)^i \binom{m}{i} (m-i)^n$.

SHORT DERIVATION

Surjections by inclusion–exclusion

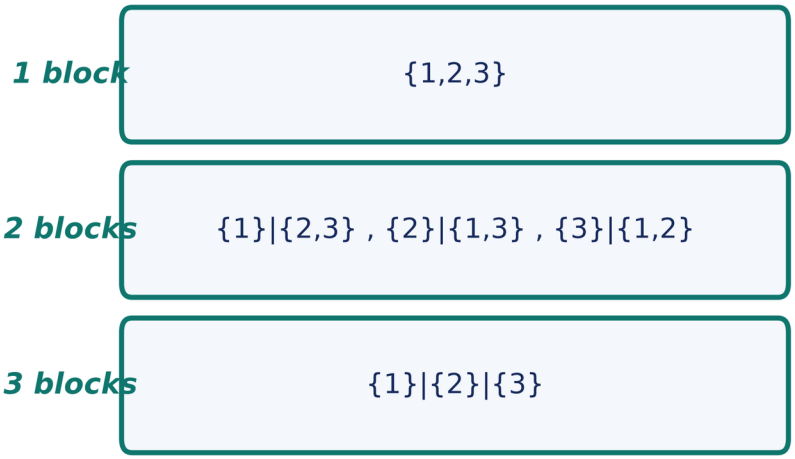
$f : A \to B$ surjective iff no element of B is missed. Let U = all m^n functions and E_b = functions missing the value b. We want $|U \setminus \bigcup_b E_b|$. Now $|E_b| = (m-1)^n$ (values chosen from the rest), $|E_b \cap E_c| = (m-2)^n$, and so on, so inclusion–exclusion gives $\sum_{i=0}^{m} (-1)^i \binom{m}{i} (m-i)^n$. Small case $n = 3, m = 2$: $2^3 - 2 \cdot 1^3 = 6$ ✓ matches the 6 onto functions $\{1,2,3\} \to \{a,b\}$ found by cases in Part I. Sanity: $m > n$ gives 0 (e.g. $n = 2, m = 3$: $9 - 3 \cdot 4 + 3 \cdot 1 = 0$).

WORKED EXAMPLE

Worked. Onto functions $\{1,2,3,4\} \to \{a,b,c\}$: $3^4 - \binom{3}{1}2^4 + \binom{3}{2}1^4 - \binom{3}{3}0^4 = 81 - 48 + 3 = 36$. Cross-check with Stirling: $S(4,3) = 6$ (partitions of 4 elements into 3 blocks: one block of 2 and two singletons, $\binom{4}{2} = 6$), then order the blocks onto a, b, c: $6 \cdot 3! = 36$ ✓. Two independent methods agreeing is the standard olympiad "check" line.

Bell numbers by listings (do this in solutions)

For $A = \{1, 2, 3, 4\}$, $B_4 = 15$ partitions: 1 block (1 way), 2 blocks: choose the block containing 1 ($S(4, 2) = 7$), 3 blocks ($S(4, 3) = 6$), 4 singletons (1). Total $1 + 7 + 6 + 1 = 15$ ✓. General recurrence (extension): $B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_k$ — element $n + 1$ joins a block after k others are already placed... more cleanly: $B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_{n-k}$: choose which elements share the block of $n + 1$.



The 5 equivalence relations on {1,2,3} as partitions (Bell B_3 = 5)

COMMON TRAP

- **Labelled vs unlabelled.** $\{\{1, 2\}, \{3, 4\}\}$ and $\{\{1, 3\}, \{2, 4\}\}$ are different partitions of a labelled set; counting unlabeled block shapes is a different (smaller) problem. State which one you count.
- Antisymmetric count $2^n 3^{n(n-1)/2}$: each off-diagonal unordered pair $\{a, b\}$ allows exactly 3 states — neither direction, only $a \to b$, only $b \to a$ — never 4 (both directions are banned).
- Inclusion–exclusion signs alternate. Boundary convention: a function on an empty domain is unique (the empty function), so $m^0 = 1$ for every m, including $0^0 = 1$ — state the convention whenever $n = 0$ appears.
- Surjections need $n \geq m$; injections need $n \leq m$. A formula that spits out a nonzero answer violating these is being used wrong.

ASSUMPTIONS & CONVENTIONS

Counts assume **finite** sets and labelled elements throughout. $S(n, m)$ here means Stirling numbers of the second kind (onto functions divided by $m!$); the notation is Part II vocabulary, but every *numerical* claim above is checkable by listing for $n, m \le 4$ — an olympiad write-up should always include one such listing.

LESSON 02 / 03

Status of this sheet

The equations below are **additional theory** (classic RMO/INMO and AIME-adjacent material). Nothing here contradicts the syllabus: every model shown is solvable with the Part I supply list plus one extra idea, stated openly. Do not claim such an equation "can never appear in JEE" — exam papers have historically leaked olympiad-flavoured models; what is not promised is any theorem beyond the statements given here.

The Cauchy equation $f(x+y)=f(x)+f(y)$ on $\mathbb{R}$

SHORT DERIVATION

Rational linearity

$f(0)=f(0+0)=f(0)+f(0)\Rightarrow f(0)=0$. Induction: $f(nx)=nf(x)$ for $n\in\mathbb{N}$; then $0=f(0)=f(x+(-x))=f(x)+f(-x)\Rightarrow f(-x)=-f(x)$, so $f(nx)=nf(x)$ for all integers n. For p/q with $q\geq 1$: $f(p)=f\left(q\cdot\frac{p}{q}\right)=qf\left(\frac{p}{q}\right)\Rightarrow f\left(\frac{p}{q}\right)=\frac{p}{q}f(1)$. Hence $f(r)=cr$ for every rational r, where $c=f(1)$.

KEY RESULT

Regular upgrade. If f is additionally monotone on any interval, or continuous at a point, or bounded above/below on any interval, then $f(x)=cx$ for all real x (rational-linearity plus density of $\mathbb{Q}$ and the squeeze). Without some regularity, non-linear solutions exist (via a Hamel basis; they are non-measurable and cannot be written down explicitly) — so: **never write $f(x)=cx$ as "the" solution unless the problem grants a regularity condition.** Many exam models quietly assume continuity; state it.

WORKED EXAMPLE

Worked models.

1. $f(x+y)=f(x)+f(y)$ for all $x,y\in\mathbb{Q}$, $f(1)=2$: then $f(7/3)=\frac{7}{3}\cdot 2=\frac{14}{3}$ — no regularity needed, because the domain is $\mathbb{Q}$ itself (rational linearity already complete).
2. $f:\mathbb{R}\to\mathbb{R}$ continuous, $f(x+y)=f(x)+f(y)$, $f(2)=6$: $c=f(1)=3$, so $f(x)=3x$ for every real x. **Check:** $f(x+y)=3(x+y)=3x+3y$ ✓.
3. $f:\mathbb{R}\to\mathbb{R}$ with $f(x+y)=f(x)+f(y)$ and $f(x)\geq 0$ for all $x>0$ (bounded below on $(0,\infty)$): the regularity condition is what upgrades $f(r)=cr$ to $f(x)=cx$; then $f(x)\geq 0$ for $x>0$ forces $c\geq 0$.

The multiplicative cousins

KEY RESULT

$f(x+y)=f(x)f(y)$: either $f\equiv 0$, or $f(0)=1$ and f never vanishes; with continuity $f(x)=a^x$, $a=f(1)>0$. $f(xy)=f(x)+f(y)$ on $(0,\infty)$: with continuity $f(x)=c\ln x$ (set $g(t)=f(e^t)$ and Cauchy reduces). $f(xy)=f(x)f(y)$ on $(0,\infty)$: $f(x)=x^c$ or $f\equiv 0,1$ specials. **Pexider:** $f(x+y)=g(x)+h(y)$ for all x,y implies all three are affine with $f=g+h+\text{const}$ relationships — plug $y=0$ to express g,h through one function and fall back to Cauchy. **Jensen:** $f\left(\frac{x+y}{2}\right)=\frac{f(x)+f(y)}{2}$ with continuity $\Rightarrow f(x)=cx+d$.

SHORT DERIVATION

$f(x+y)=f(x)f(y)$ quickly

Put $y=0$: $f(x)=f(x)f(0)$, so either $f\equiv 0$ or $f(0)=1$. If $f(x_0)=0$ for some x_0, then $f(x)=f((x-x_0)+x_0)=f(x-x_0)f(x_0)=0$ for all x: $f\equiv 0$. So a non-zero solution never vanishes; if it takes both signs it still can't cross 0 (continuous image of connected set), hence a positive solution has $f(x)=e^{g(x)}$ with g additive, and continuity finishes $g(x)=x\ln a$.

Involutions $f(f(x))=x$

$f\circ f=\mathrm{id}$ means f is its own inverse: f is bijective and $f^{-1}=f$. Solutions of $f(f(x))=x$ split into **fixed points** ($f(x)=x$) and **2-cycles** ($f(x)=y\neq x$, $f(y)=x$).

SHORT DERIVATION

Increasing involution fixes everything

Suppose f is strictly increasing and $f(f(x))=x$. If $f(x_0)>x_0$ then applying the increasing f gives $f(f(x_0))>f(x_0)>x_0$, i.e. $x_0>f(x_0)$ — contradiction. Similarly $f(x_0)<x_0$ contradicts. So $f(x)=x$ for all x: every increasing involution is the identity on its domain. A decreasing involution (e.g. $f(x)=a-x$, $f(x)=1/x$ on $\mathbb{R}^+$) instead swaps pairs around fixed points.

WORKED EXAMPLE

Worked. Find all continuous $f:\mathbb{R}\to\mathbb{R}$ with $f(f(x))=x$ and f decreasing. Any x with $f(x)=x$ is a fixed point; $g(x)=f(x)-x$ is decreasing (f dec, $-x$ dec... sum of two decreasing is decreasing) hence g has at most one zero — at most one fixed point. Example $f(x)=a-x$ has exactly one ($x=a/2$). The general decreasing-continuous involution picture: one fixed point (if g changes sign) and the rest of the line paired off in 2-cycles symmetric about it. **Check:** $f(x)=1/x$ on $(0,\infty)$: decreasing, $f(f(x))=x$ ✓, fixed point $x=1$ ✓.

COMMON TRAP

- Cauchy with domain $\mathbb{Q}$: rational linearity is the complete answer. Cauchy with domain $\mathbb{R}$: rational linearity + "regularity" is required; state the condition.
- $f(x) = x^2$ is even, $f(x) = x^3$ odd, but $f(x) = x^2 + x^3$ is neither — decomposition (Part I) is the safe tool.
- In $f(f(x)) = x$ problems, plugging $x = f(t)$ is the standard trick: $f(f(f(t))) = f(t)$, i.e. $f^3 = f$; combined with $f^2 = \mathrm{id}$ this just re-proves bijectivity — write the injection/surjection argument once, cleanly.
- Periodic + additive: $f(x + T) = f(x)$ with additivity forces $f(T) = f(0) = 0$, and with continuity that collapses $f \equiv 0$ (since $f(r) = cr$ and $cT = 0$). Without continuity a wild additive T-periodic solution can still exist on a Hamel basis — so again: state the regularity condition before claiming collapse. The point stands: ***imposing two strong structures usually collapses the family to trivial members, but "usually" needs a hypothesis.***

LESSON 03 / 03

Status of this sheet

The greatest-integer function is **in the official syllabus** (listed under special functions); Hermite's identity and the fixed-point decomposition are extra theory. Every worked model below is solved with syllabus tools only — so treat the identities as accelerators, and the models as fair game.

Hermite's identity

SHORT DERIVATION

Sum of n shifted floors

For real x and positive integer n:

$$\sum_{k=0}^{n-1} \lfloor x + \tfrac{k}{n} \rfloor = \lfloor nx \rfloor$$

Write $x = m + \theta$ with $m = \lfloor x \rfloor$, $\theta \in [0, 1)$. Then $\lfloor x + \frac{k}{n} \rfloor = m + \lfloor \theta + \frac{k}{n} \rfloor$, so the left side is $nm + \sum_k \lfloor \theta + \frac{k}{n} \rfloor$. Now $\lfloor \theta + \frac{k}{n} \rfloor = 1$ exactly when $\frac{k}{n} \geq 1 - \theta$, i.e. $k \geq n(1-\theta)$; the number of $k \in \{0, \ldots, n-1\}$ with $k \geq n(1-\theta)$ is $\lfloor n\theta \rfloor$ (boundary $n(1-\theta)$ an integer counts k from that value up). So the sum $= nm + \lfloor n\theta \rfloor = \lfloor nm + n\theta \rfloor = \lfloor nx \rfloor$ ✓.

WORKED EXAMPLE

Worked checks. $n = 3$, $x = 1.7$:
$\lfloor 1.7 \rfloor + \lfloor 1.7 + 0.33 \rfloor + \lfloor 1.7 + 0.67 \rfloor = 1 + 2 + 2 = 5 = \lfloor 5.1 \rfloor$ ✓. $x = 2$ exactly:
$2 + 2 + 2 = 6 = \lfloor 6 \rfloor$ ✓. **Check boundary:** when $n\theta$ is an integer the threshold $k \geq n(1-\theta)$ is attained (included) — $\lfloor \cdot \rfloor$ inequality is $\geq$ on both sides of the threshold count.

Floor identities and equations

KEY RESULT

$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$; $\lfloor x \rfloor + \lfloor y \rfloor \leq \lfloor x + y \rfloor \leq \lfloor x \rfloor + \lfloor y \rfloor + 1$; $\lfloor x \rfloor + \lfloor -x \rfloor = 0$ if $x \in \mathbb{Z}$ else -1; $\{x\} + \{-x\} = 0$ or 1 with the same split; number of integers in $(a, b]$ is $\lfloor b \rfloor - \lfloor a \rfloor$; $\lfloor x \rfloor = c \Leftrightarrow x \in [c, c+1)$; $\lfloor x \rfloor < c \Leftrightarrow x < c$ (for integer c); $\lfloor x \rfloor \geq c \Leftrightarrow x \geq c$ (integer c).

STANDARD APPLICATION

Model (fully syllabus-level). Solve $\lfloor x^2 \rfloor = 2$ over the reals. Sandwich: $2 \le x^2 < 3$. Left half: $x^2 \ge 2 \Rightarrow |x| \ge \sqrt{2}$; right half: $x^2 < 3 \Rightarrow |x| < \sqrt{3}$. Solution set $= \{x : \sqrt{2} \le |x| < \sqrt{3}\} = [-\sqrt{3}, -\sqrt{2}] \cup [\sqrt{2}, \sqrt{3})$ — **closed** at $\pm\sqrt{2}$ ($x^2 = 2$ gives $\lfloor 2 \rfloor = 2$ ✓) and **open** at $\pm\sqrt{3}$ ($x^2 = 3$ would give $\lfloor 3 \rfloor = 3$ ✗). The answer is a set of reals, not a number. Integer version: $x \in \mathbb{Z}$ needs $x^2 \in [2, 3)$, but squares of integers are $0, 1, 4, \ldots$ — **no solutions**. **Check:** the two phrasings ("all real x" vs "all integer x") have completely different answers; the quantifier is part of the question.

Fixed points and involutions (counting flavour)

$f(x) = x$ is a **fixed point**; $f(f(x)) = x$ is an **involution equation** whose solution set decomposes into fixed points of f plus 2-cycles $\{x, f(x)\}$ with $f(x) \neq x$.

KEY RESULT

If f is strictly increasing and $f(f(x)) = x$ on an interval, then $f(x) = x$ everywhere on it (Part II sheet 02 proof). If f is strictly decreasing and continuous, $g(x) = f(x) - x$ is strictly decreasing, so f has **at most one** fixed point; each non-fixed x pairs with $f(x)$ in a 2-cycle. Counting solutions of $f(f(x)) = x$ for a concrete monotone f therefore reduces to: find fixed points (solve $f(x) = x$), then account for the symmetric 2-cycles.

WORKED EXAMPLE

Worked. $f(x) = 3 - x$ on $\mathbb{R}$: decreasing, $f(f(x)) = 3 - (3 - x) = x$ ✓ for **every** x — the entire line solves the involution equation, yet only $x = 3/2$ is a fixed point. Moral: "solutions of $f(f(x)) = x$" and "fixed points of f" are different sets; the question wording decides which is asked. With $f(x) = \frac{1}{x}$ on $(0, \infty)$: $f(f(x)) = x$ everywhere; fixed point $x = 1$ only. **Check:** $f(2) = 1/2$, $f(1/2) = 2$ — a 2-cycle, as predicted by the decreasing-case analysis.

COMMON TRAP

- $\lfloor x \rfloor$ of a negative number: $\lfloor -1.2 \rfloor = -2$, not -1 (greatest integer **less or equal**). Truncation toward zero is a different function.
- In floor equations, the answer set's endpoints are decided by strict vs non-strict inequalities — write the sandwich first, then convert to interval notation, then test both endpoints.
- $\lfloor x^2 \rfloor = 2$ vs $\lfloor x \rfloor^2 = 2$: the second means $\lfloor x \rfloor = \pm\sqrt{2}$, impossible for integer $\lfloor x \rfloor$ — no solutions. Composition order matters even inside floor equations.
- Never square an inequality $\sqrt{2} \le |x| < \sqrt{3}$ without tracking that $|x|$ is already non-negative — squaring is safe on $[0, \infty)$ but flips nothing here; for $x < 0$ reflect by $|x|$.

ASSUMPTIONS & CONVENTIONS

All floor statements are for real x; n is a positive integer in Hermite; fixed-point claims assume the stated monotonicity/continuity on an interval. 2-cycles are unordered pairs $\{x, y\}$ with $y = f(x) \neq x$; when "count solutions" is asked on a finite set, count elements x, not cycles.